

Chapter 108

Mask Nodes

This chapter details the Mask nodes available in Fusion.

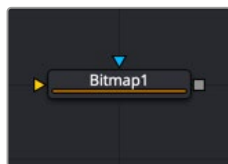
The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Bitmap Mask [Bmp]



The Bitmap node

Bitmap Mask Node Introduction

The Bitmap mask node allows images from the node tree to act as masks for nodes and effects. Bitmap masks can be based on values from any of the color, Alpha, hue, saturation, luminance, and auxiliary coverage channels of the image. Nodes can also be masked based on the Object ID or Material ID of a 3D-rendered image (provided those channels were included when the file was rendered).

The Bitmap mask node is not required for effect masks. For effects masks, the Common Settings tab for the masked node displays controls to select which channel of the mask image is used to create the mask.

However, Bitmap mask nodes may still be required to connect to other mask inputs on some nodes, such as Garbage Mattes and Pre-Masks. Also, using a Bitmap mask node between the mask source and the target node provides additional options that would not be available when connecting directly, such as combining masks, blurring the mask, or clipping its threshold.

Inputs

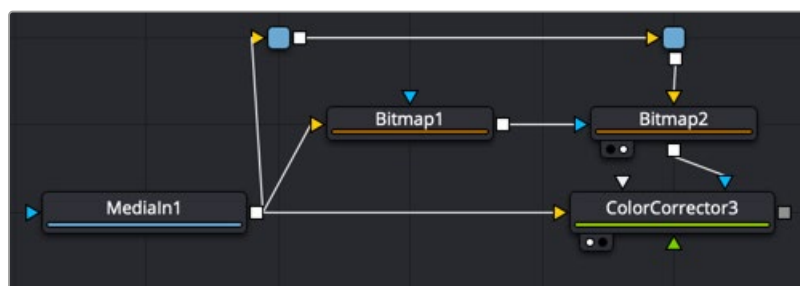
The Bitmap mask node includes two inputs in the Node Editor.

Input: The orange input accepts a 2D image from which the mask will be created.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

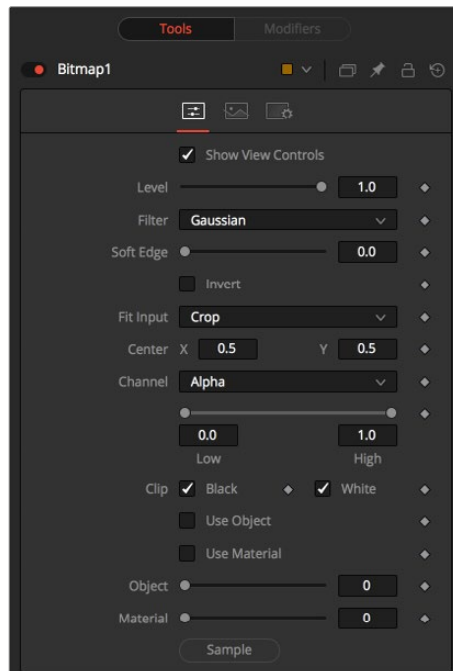
Basic Node Setup

The Bitmap mask node is not required for connecting an image into the effect mask input, but it does provide options that are otherwise unavailable. It allows for selecting channels other than RGBA for the mask, as well as softness and clipping. In the node tree below, the two Bitmap masks are combined using a Paint menu located in the second Bitmap mask node, which allows you to add, subtract, multiply, and perform other operations on the combined mask.



Bitmap nodes can be chained together for more advanced matte operations.

Inspector



Bitmap Mask controls

Controls Tab

The Controls tab is used to refine how the image connected to the orange input converts into the Bitmap mask.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the Blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask’s values add to the input mask’s values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask’s values.
- **Minimum:** Comparing the input mask’s values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask’s values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask’s values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, this checkbox affects all pixels, regardless of whether the new mask covers them.

Fit Input

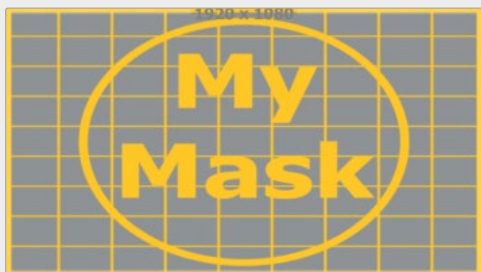
This menu is used to select how the image source is treated if it does not fit the dimensions of the generated mask.

In the example below, a 720 x 576 image source (yellow) is used to generate a 1920 x 1080 mask (gray).

Crop: If the image source is smaller than the generated mask, it will be placed according to the X/Y controls, masking off only a portion of the mask. If the image source is larger than the generated mask, it will be placed according to the X/Y controls and cropped off at the borders of the mask.



Stretch: The image source will be stretched in X and Y to accommodate the full dimensions of the generated mask. This might lead to visible distortions of the image source.



Inside: The image source will be scaled uniformly until one of its dimensions (X or Y) fits the inside dimensions of the mask. Depending on the relative dimensions of the image source and mask background, either the image source's width or height may be cropped to fit the respective dimensions of the mask.



Width: The image source will be scaled uniformly until its width (X) fits the width of the mask. Depending on the relative dimensions of the image source and mask, the image source's Y dimension might not fit the mask's Y dimension, resulting in either cropping of the image source in Y or the image source not covering the mask's height entirely.



Height: The image source will be scaled uniformly until its height (Y) fits the height of the mask. Depending on the relative dimensions of the image source and mask, the image source's X-dimension might not fit the mask's X-dimension, resulting in either cropping of the image source in X or the image source not covering the mask's width entirely.



Outside: The image source will be scaled uniformly until one of its dimensions (X or Y) fits the outside dimensions of the mask. Depending on the relative dimensions of the image source and mask, either the image source's width or height may be cropped or not fit the respective dimension of the mask.



Center X and Y

These controls adjust the position of the Bitmap mask.

Channel

The Channel menu determines the Channel of the input image used to create the mask. Choices include the red, green, blue, and alpha channels, the hue, luminance, or saturation values, or the auxiliary coverage channel of the input image (if one is provided).

Threshold Low/High

The Threshold range control can be used to clip the bitmap image. Increasing the low range control will clip pixels below the specified value to black (0.0). Decreasing the high range control will force pixels higher than the specified value to white (1.0).

Use Object/Use Material

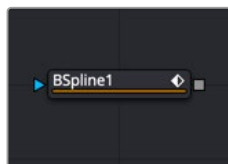
This control has no effect unless the input image contains a Material or Object ID channel. When toggled on, the Object ID and Material ID are used to create a mask based on the selected object or material. When toggled off, the regular color channels will generate the mask.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are also duplicated in other Mask nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

B-Spline Mask [BSp]



The B-Spline node

A B-Spline mask is identical to a Polygon mask in all respects except one. Where Polygon masks use Bézier splines, this mask node uses B-Splines. Where Bézier splines employ a central point and two handles to manage the smoothing of the spline segment, a B-Spline requires only a single point. This means that a B-Spline shape requires far fewer control points to create a nicely smoothed shape.

When first added to a node, the B-Spline mask consists of only Center control, which is visible onscreen. Points are added to the B-Spline by clicking in the viewer. Each new point is connected to the last one created, but instead of the spline going directly through each control point, B-Spline control points only influence the spline shape. The control point pulls the spline in its direction to create a smooth curve.

Like the Polygon mask tool, the B-Spline mask auto-animates. Adding this node to the Node Editor adds a keyframe to the current frame. Moving to a new frame and changing the shape creates a new keyframe and interpolate between the two defined shapes.

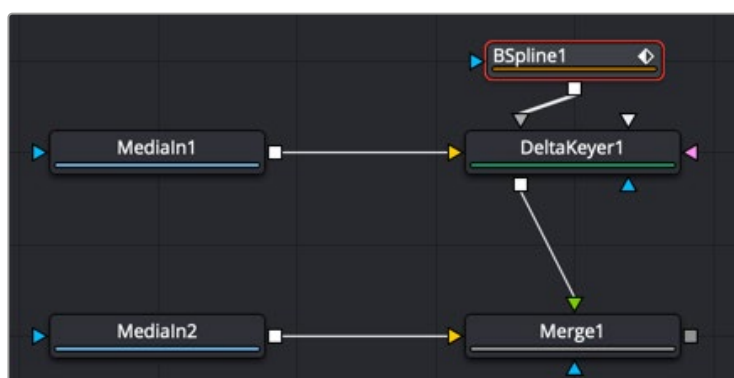
Inputs

The B-Spline mask node includes a single effect mask input.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

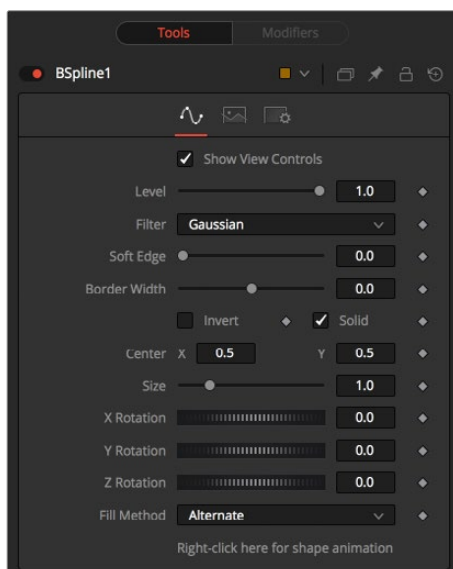
Basic Node Setup

The B-Spline node can be used to generate a single smooth spline shape or combined with other masks for more complex shapes. In the node tree below, the B-Spline mask is used to generate a smooth, curved shape as a garbage matte on the Delta Keyer.



A B-Spline node generates a smooth, curved shape as a garbage matte

Inspector



B-Spline Mask controls

Controls Tab

The Controls tab is used to refine how the B-Spline appears after drawing it in the viewer.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all of the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.

- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Border Width

The Border Width control adjusts the thickness of the mask’s edge. When the solid checkbox is toggled on, the border thickens or narrows the mask. When the mask is not solid, the mask shape draws as an outline, and the width uses the Border Width setting.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask’s values add to the input mask’s values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask’s values.
- **Minimum:** Comparing the input mask’s values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask’s values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask’s values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, the checkbox affects all pixels, regardless of whether the new mask covers them or not.

Solid

When the Solid checkbox is enabled, the mask is filled to be transparent (white) unless inverted. When disabled, the spline is drawn as just an outline whose thickness is determined by the Border Width slider.

Center X and Y

These controls adjust the position of the B-Spline mask.

Size

Use the Size control to adjust the scale of the B-Spline effect mask, without affecting the relative behavior of the points that compose the mask or setting a keyframe in the mask animation.

X, Y, and Z Rotation

Use these three controls to adjust the rotation angle of the mask along any axis.

Fill Method

The Fill Method menu offers two different techniques for dealing with overlapping regions of a polyline. If overlapping segments in a mask are causing undesirable holes to appear, try switching the setting of this control from Alternate to Non Zero Winding.

Right-Click Here for Shape Animation

By default, all B-Spline masks are animated when they are created. The initial keyframe is set to the current time, and any changes to the shape at different times will create new keys.

Right-clicking on this label will display a contextual menu that offers options for removing or re-adding animation to the mask, or publishing and connecting the masks.

Adding Points

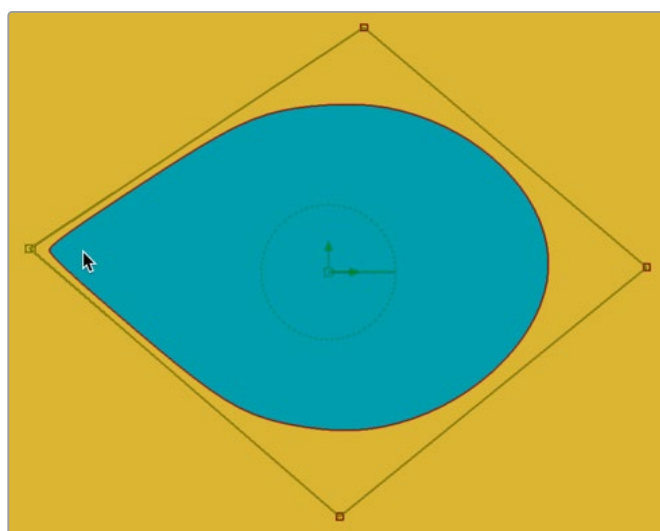
Adding Points to a B-Spline effect mask is relatively simple. Immediately after adding the node to the Node Editor, there are no points, but the tool will be in Click Append mode. Click once in the viewer wherever a point is required for the mask. Continue clicking to draw the shape of the mask.

When the shape is complete, click on the initial point again to close the mask.

When the shape is closed, the mode of the polyline changes to Insert and Modify. This allows you to add and adjust additional points on the mask by clicking the spline segments. To lock down the mask's shape and prevent accidental changes, switch the Polyline mode to Done using the Polyline toolbar or contextual menu.

Adjusting Tension on a B-Spline

The tension of the control point determines the smoothness of a B-Spline. To adjust the tension of a B-Spline's control points, select the point in the viewer, hold down the W key and drag the mouse pointer to the left and right to increase or decrease the tension of the curve through that point.



Adjusting Tension on a B-Spline

B-Spline Toolbar

When a B-Spline mask is selected in the Node Editor, a toolbar appears above the viewer with buttons for easy access to the modes. Position the pointer over any button in the toolbar to display a tooltip that describes that button's function.



B-Spline Mask Polygon toolbar

You can change the way the toolbar is displayed by right-clicking on the toolbar and selecting from the options displayed in the toolbar's contextual menu.

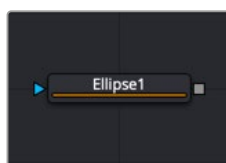
The functions of the buttons in this toolbar are explained in depth in the Polylines section.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Ellipse Mask [Elp]



The Ellipse node

Ellipse Mask Node Introduction

The Ellipse mask is most useful for masking round objects. It is a circle by default, but independent control is offered over the width, height, and angle, providing for a wide variety of ellipsoidal shapes.

Inputs

The Ellipse mask node includes a single effect mask input.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

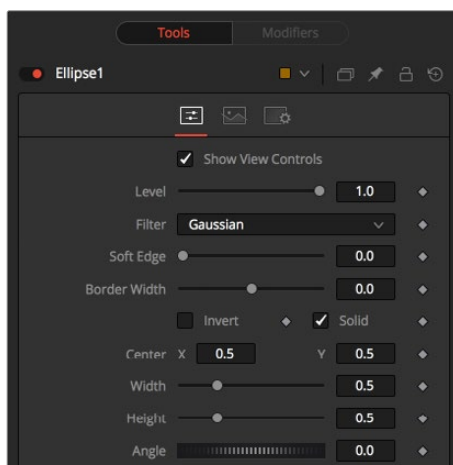
Basic Node Setup

The Ellipse mask node is useful for generating circular masks. Below, the Ellipse mask is used to generate a circular colored shape by cutting a circular shape from a background node.



Ellipse mask node creating a colored circle by masking a background node

Inspector



Ellipse Mask controls

Controls Tab

The Controls tab is used to refine how the ellipse appears after drawing it in the viewer.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all of the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Border Width

The Border Width control adjusts the thickness of the mask's edge. When the solid checkbox is toggled on, the border thickens or narrows the mask. When the mask is not solid, the mask shape draws as an outline, and the width uses the Border Width setting.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask's values add to the input mask's values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask's values.
- **Minimum:** Comparing the input mask's values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask's values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask's values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, the checkbox affects all pixels, regardless of whether the new mask covers them or not.

Solid

When the Solid checkbox is enabled, the mask is filled to be transparent (white) unless inverted. When disabled, the spline is drawn as just an outline whose thickness is determined by the Border Width slider.

Center X and Y

These controls adjust the position of the Ellipse mask.

Width

This control allows independent control of the ellipse mask's Width. In addition to the slider in the mask's controls, interactively drag the width (left or right edge) of the mask on the viewer using the pointer. Any changes will be reflected in this control.

Height

Height allows independent control of the ellipse mask's height. In addition to the slider in the mask's controls, interactively drag the height (top or bottom edge) of the mask on the view using the pointer. Any changes will be reflected in this control.

To change the mask's size without affecting the aspect ratio, drag the onscreen control between the edges (diagonal). This will modify both the width and height proportionately.

Angle

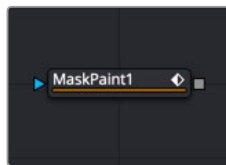
Change the rotational angle of the mask by moving the Angle control left or right. Values can be entered into the number fields provided. Alternately, use the onscreen controls by dragging the little circle at the end of the dashed angle line to interactively adjust the rotation of the ellipse.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Mask Paint [PnM]



The Mask Paint node

Mask Paint Node Introduction

The Mask Paint node allows direct painting of mask images, using the pointer as if it was a paintbrush. In addition to regular paint strokes, it is possible to apply basic primitive shapes and polyline style strokes.

Each stroke can have a duration that lasts for the entire project, a single frame, or an arbitrary number of fields. The strokes can have independent durations in the Keyframes Editor for easy manipulation of time. Alternatively, Multistrokes is a faster but non-editable way for doing many mask clean up paint tasks.

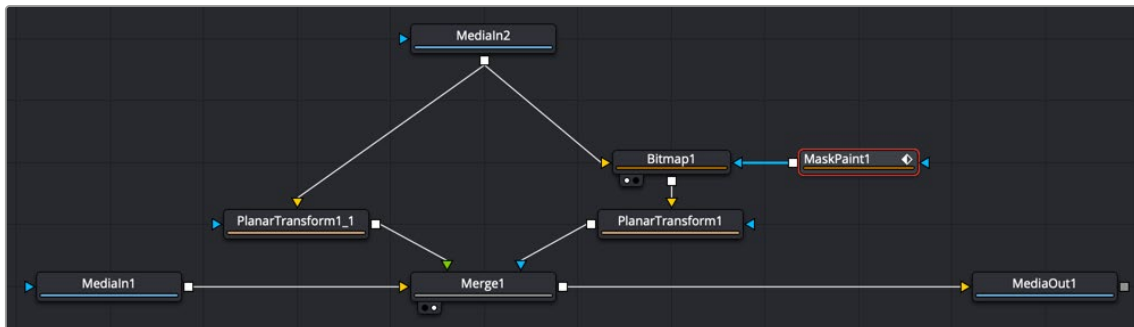
Inputs

The Paint mask node includes a single effect mask input.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

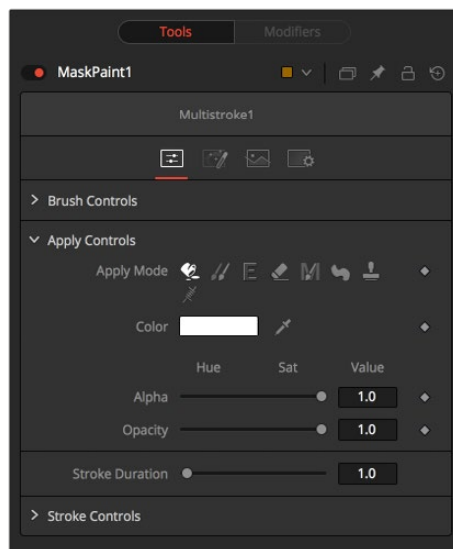
Basic Node Setup

The Mask Paint node is useful for painting masks using a more free hand, pressure sensitive style. In the node tree below, the Mask Paint node is used to patch up holes in a Bitmap mask.



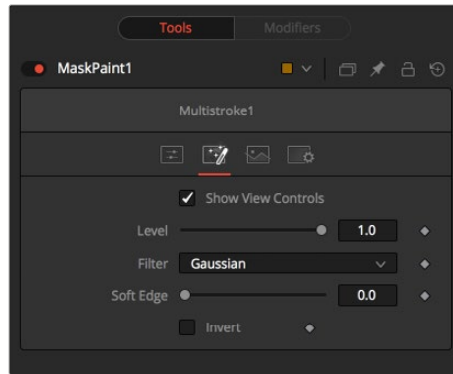
A Mask Paint node can be used to repair problematic areas of a matte.

Inspector



Mask Paint controls

As the Controls tab in the Mask Paint node is fundamentally identical to the Paint node, for more detail, see *Chapter 113, "Paint Node,"* in the DaVinci Resolve Reference Manual, or Chapter 53 in the Fusion Reference Manual. The only difference between the two nodes is that, as Mask Paint operates on single-channel mask images, there is no Channel Selector control, and all color controls have only a single Alpha value. The Mask tab, however, includes several parameters that are different from the Paint tool, so they are covered below.



The Mask Paint tab

Mask Tab

The Mask tab is used to refine the basic mask parameters that do not fall into the category of “panting.” These include how multiple masks are combined, overall softness control, and level control.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the masks onscreen controls in the viewer. Onscreen controls including, center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all of the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask's values add to the input mask's values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask's values.
- **Minimum:** Comparing the input mask's values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask's values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask's values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, the checkbox affects all pixels, regardless of whether the new mask covers them or not.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

MultiPoly [MPly]



The MultiPoly node

Introduction

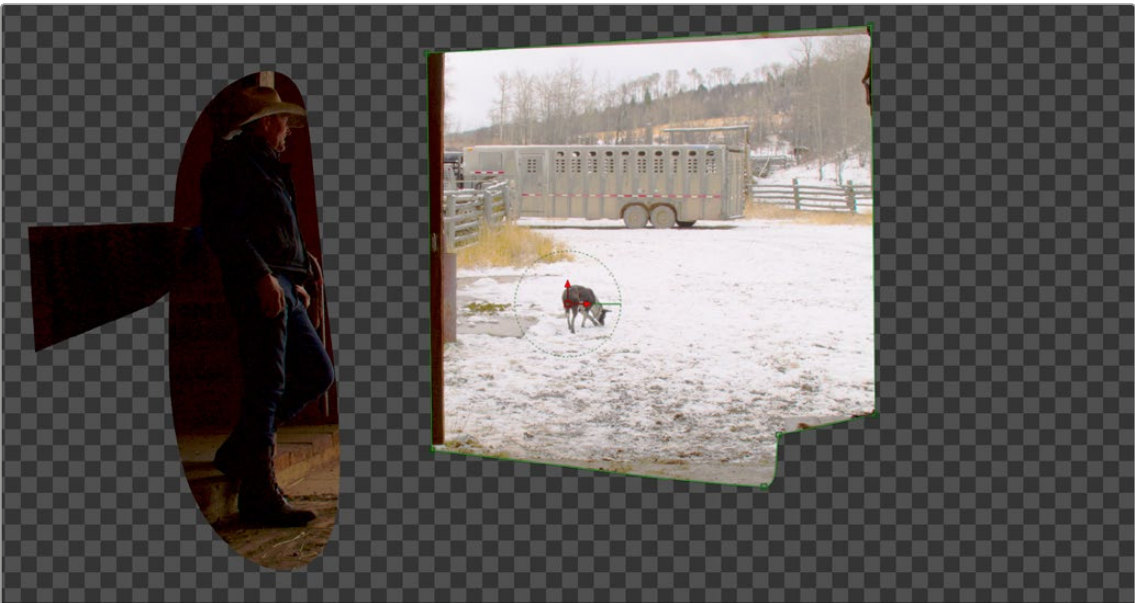
The MultiPoly tool makes rotoscoping more efficient by allowing you to create multiple masks in a single node. The list of polygon shapes in the inspector allows you to easily modify, organize, and animate shapes through one simple interface.

With the multiple masks presented in a scrolling list format, akin to the convenience offered by MultiMerge, you can effortlessly organize, edit, and animate individual shapes. This intuitive interface streamlines the process of drawing intricate masks, empowering you to create polished compositions with ease.

Furthermore, all these advantages are available within this List view, facilitating swift selection and modification of specific shapes, toggling their visibility, and precise adjustment of their parameters.

Usage

After adding the MultiPoly, the tool is ready for you to start drawing your first mask. To create a mask, use the tools at the top of the viewer to create polygons, splines, and points (these points will make up segments of the polyline). You can close the mask by clicking the initial point again.



The MutliPoly tool showing three creation of masks on the viewer

Inspector



Multipoly controls

In the Inspector, pressing the Polygon or BSpline button will create a new mask type. Each new mask is added to the List view located in the Inspector.

Within the List view, you can rename or reorder any of the masks you've just added. Modify specific shapes by selecting them in the layer list and changing their properties in the lower portion of the Inspector.

To animate any of the drawn masks, select the animation diamond next to the text "Right-click here for shape animation," at the bottom of the Inspector or you can right-click on this text and select Animate.

Within the layer list, right clicking on a mask offers a few options.

Duplicate: Makes a copy of the selected mask and adds a new entry in the layer list.

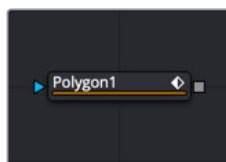
Split here: Splits the selected mask and any mask below it, copying them to a new fresh MultiPoly node.

Rename: Allows you to rename the selected mask.

Reset to default: Keeps the mask entry in the layer list but removes the mask shape and resets all the controls for that mask.

Delete: Deletes the mask.

Polygon Mask [Ply]



The Polygon node

Polygon Mask Node Introduction

The Polygon mask is most useful for masking objects that do not have a regular shape. When first added to a node, the Polygon mask consists of only Center and Angle controls, which are visible onscreen. Points are added to the polyline by clicking in the viewer. Each new point is connected to the last one created.

Like the B-Spline mask tool, the Polygon mask auto-animates. Adding this node to the Node Editor adds a keyframe to the current frame. Moving to a new frame and changing the shape creates a new keyframe and interpolate between the two defined shapes.

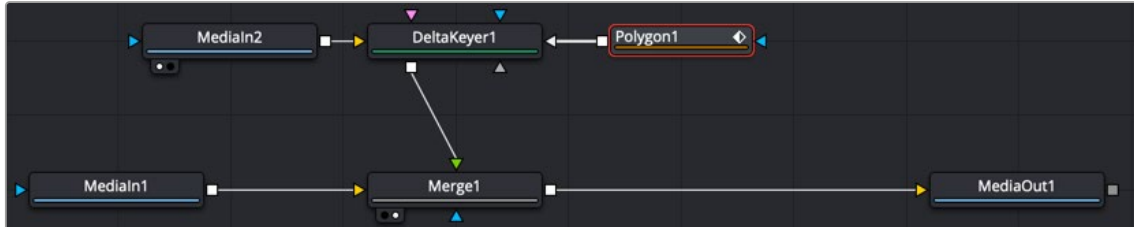
Inputs

The Polygon mask node includes a single effect mask input.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

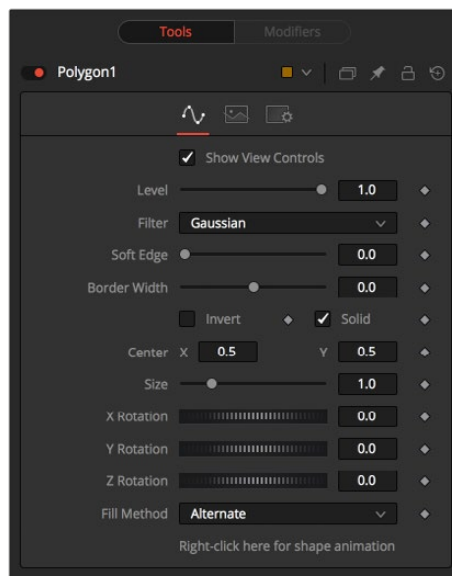
Basic Node Setup

The Polygon node can be used to generate a detailed spline shape or combined with other masks for even more complex shapes. In the node tree below, the Polygon mask is used to generate detailed shape as a solid matte on the Delta Keyer.



A Polygon node generates a detailed shape as a Solid matte.

Inspector



Polygon Mask controls

Controls Tab

The Controls tab is used to refine how the polyline appears after drawing it in the viewer.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all of the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Border Width

The Border Width control adjusts the thickness of the mask’s edge. When the solid checkbox is toggled on, the border thickens or narrows the mask. When the mask is not solid, the mask shape draws as an outline, and the width uses the Border Width setting.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask’s values add to the input mask’s values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask’s values.
- **Minimum:** Comparing the input mask’s values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask’s values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask’s values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, the checkbox affects all pixels, regardless of whether the new mask covers them or not.

Solid

When the Solid checkbox is enabled, the mask is filled to be transparent (white) unless inverted. When disabled, the spline is drawn as just an outline whose thickness is determined by the Border Width slider.

Center X and Y

These controls adjust the position of the polygon spline mask.

Size

Use the Size control to adjust the scale of the polygon spline effect mask, without affecting the relative behavior of the points that compose the mask or setting a keyframe in the mask animation.

X, Y, and Z Rotation

Use these three controls to adjust the rotation angle of the mask along any axis.

Fill Method

The Fill Method menu offers two different techniques for dealing with overlapping regions of a polyline. If overlapping segments in a mask are causing undesirable holes to appear, try switching the setting of this control from Alternate to Non Zero Winding.

Right-Click Here for Shape Animation

By default, all polygon spline masks are animated when they are created. The initial keyframe is set to the current time, and any changes to the shape at different times create new keys.

Right-clicking on this label displays a contextual menu that offers options for removing or re-adding animation to the mask, or publishing and connecting the masks together.

Common Controls**Image and Settings Tabs**

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Adding Points

Adding Points to a polygonal effect mask is relatively simple. Immediately after adding the node to the Node Editor, there are no points, but the tool will be in Click Append mode. Click once in the viewer wherever a point is required for the mask. Continue clicking to draw the shape of the mask. When the shape is complete, click on the initial point again to close the mask.

When the shape is closed, the mode of the polyline will change to Insert and Modify. This allows for the adjusting and adding of additional points to the mask by clicking on segments of the polyline. To lock down the mask’s shape and prevent accidental changes, switch the Polyline mode to Done using the Polyline toolbar or contextual menu.



B-Spline Mask Polygon toolbar

When a Polygon (or B-Spline) mask is added to a node, a toolbar appears above the viewer, offering easy access to modes. Hold the pointer over any button in the toolbar to display a tooltip that describes that button's function.

- **Click:** Click is the default option when creating a polyline (or B-Spline) mask. It is a Bézier style drawing tool. Clicking sets a control point and appends the next control point when you click again in a different location.
- **Draw:** Draw is a freehand drawing tool. It creates a mask similar to drawing with a pencil on paper. You can create a new mask using the Draw tool, or you can extend an existing open spline by clicking the Draw tool and starting to draw from the last control point.
- **Insert:** Insert adds a new control point along the spline.
- **Modify:** Modify allows you to safely move or smooth any exiting point along a spline without worrying about adding new points accidentally.
- **Done:** Prevents any point along the spline from being moved or modified. Also, new points cannot be added. You can, however, move and rotate the entire spline.
- **Closed:** Closes an open spline.
- **Smooth:** Changes the selected control point from a linear to a smooth curve.
- **Linear:** Changes the selected control point from a smooth curve to linear.
- **Select All:** Selects all the control points on the spline.
- **Keys:** Shows or hides the control points along the spline.
- **Handles:** Shows or hides the Bézier handles along the polyline.
- **Shape:** Places a reshape rectangle around the selected spline shape. Using the reshape rectangle, you can deform groups of control points or entire shapes much easier than modifying each point.
- **Delete:** Deletes the selected control point(s).
- **Reduce:** Opens a Freehand precision window that can be used to reduce the number of controls points on a spline. This can make the paint stroke easier to modify, especially if it has been created using the Draw tool.
- **Publish menu:** You can use the publish menu to select between publishing the control points or the path. Publishing is a form of parameter linking, it makes the selected item available for use by other controls. It also allows you to attach a control point to a tracker.
- **Follow Points:** Allows a selected point to follow the path of a published point. The point follows the published point using an offset position.
- **Double Poly:** Allows softening part of the spline curve while keeping other portions of the curve sharp. The double polyline is composed of two shapes, an inner and outer shape. The inner shape is the original shape from the single polyline, whereas the outer shape is used to determine the spread of the softness. The further the outer shape gets from the inner shape, the softer that segment of the shape becomes. Both polylines start with exactly the same shape as the original single polyline, keeping the mask sharp to start. Any animation already applied to the shape remains. To select the outer shape, press the Tab key to cycle between the onscreen controls until the dashed outline is visible, or you can select the outer polyline using the contextual menu's Controls > Outer Polygon menu.
- **Multiframe:** Multiframe is a method of adjusting control points across multiple keyframes. The default setting of none only adjusts the control point of a spline on the current keyframe. Setting the menu to All adjusts the controls point for all keyframes. Prev settings adjust the current and previous keyframe while Next adjusts the current and next keyframe.
- **Onion Skinning:** Enabling onion skinning displays a mix in the viewer of the spline animation. It is useful when aligning spline animation and motion. Selecting Onion Skin Settings from the drop down menu allows you to set the number of overlapping frames.

- **Roto Assist:** Enable the Roto Assist button when you begin painting with the Polyline Stroke tool. The polyline points snap to the closest edge as you click to add points to the shape. A cyan outline indicates the points that have snapped to an edge. There are three main Roto Assist options selectable through the drop down menu:

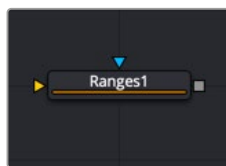
Multiple Points: When enabled, a single click on a high contrast edge adds multiple points to define the entire edge, instead of having to add each point individually. This is a one time only click. The second click reverts to single point edge detection.

Distance 8: Opens a dialog where you can set the pixel range within which searching for an edge takes place.

Reset: Used for resetting the snap attribute of all snapped points. After resetting, the points become unavailable for tracking.

Change the way the toolbar is displayed by right-clicking on the toolbar and selecting from the options displayed in the toolbar's contextual menu. The functions of the buttons in this toolbar are explained in depth in the Polylines chapter.

Ranges Mask [RNG]



The Ranges node

Ranges Mask Node Introduction

Similar to Bitmap mask, the Ranges mask allows images from the node tree to act as masks for nodes and effects. Instead of creating a simple luminance-based mask from a given channel, Ranges allows spline-based selection of low, mid and high ranges, akin to Color Corrector.

Inputs

The Ranges mask node includes two inputs in the Node Editor.

Input: The orange input accepts a 2D image from which the mask will be created.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

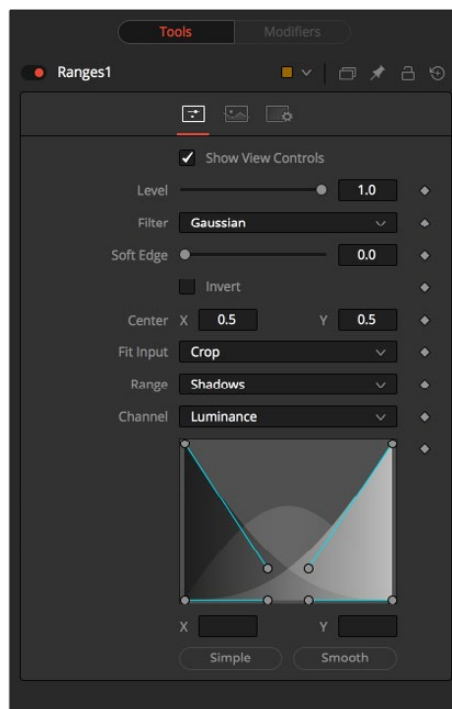
Basic Node Setup

The Ranges node is not required for connecting an image into the effect mask input, but like the Bitmap node, it does provide options that are otherwise unavailable. It allows for selecting channels other than RGBA for the mask, as well as softness and clipping. In the node tree below, the Ranges node takes the composite out of the merge, creating a mask for the color correction.



A Ranges node selects a specific range in the image to create a mask.

Inspector



Rangers Mask controls

Controls Tab

The Controls tab is used to refine how the image connected to the orange input converts into the ranges mask.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all of the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask’s values add to the input mask’s values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask’s values.
- **Minimum:** Comparing the input mask’s values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask’s values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask’s values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, the checkbox affects all pixels, regardless of whether the new mask covers them or not.

Center X and Y

These controls adjust the position of the ranges mask.

Fit Input

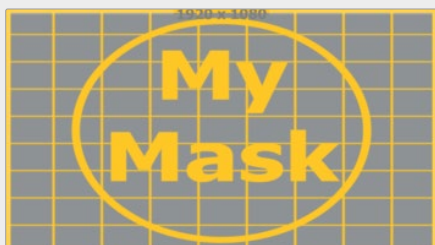
This menu is used to select how the image source is treated if it does not fit the dimensions of the generated mask.

For example, below, a 720 x 576 image source (yellow) is used to generate a 1920 x 1080 mask (gray).

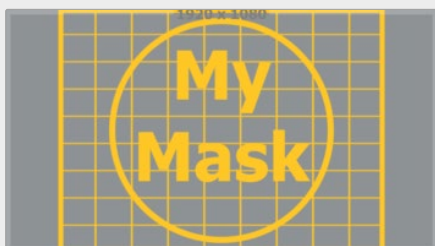
Crop: If the image source is smaller than the generated mask, it is placed according to the X/Y controls, masking off only a portion of the mask. If the image source is larger than the generated mask it is placed according to the X/Y controls and cropped off at the borders of the mask.



Stretch: The image source is stretched in X and Y to accommodate the full dimensions of the generated mask. This might lead to visible distortions of the image source.



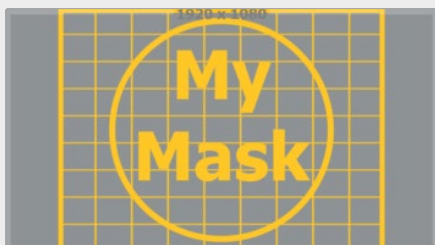
Inside: The image source is scaled uniformly until one of its dimensions (X or Y) fits the inside dimensions of the mask. Depending on the relative dimensions of the image source and mask background, either the image source's width or height may be cropped to fit the respective dimension of the mask.



Width: The image source is scaled uniformly until its width (X) fits the width of the mask. Depending on the relative dimensions of the image source and mask, the image source's Y dimension might not fit the mask's Y dimension, resulting in either cropping of the image source in Y or the image source not covering the mask's height entirely.



Height: The image source is scaled uniformly until its height (Y) fits the height of the mask. Depending on the relative dimensions of the image source and mask, the image source's X dimension might not fit the mask's X dimension, resulting in either cropping of the image source in X or the image source not covering the mask's width entirely.



Outside: The image source is scaled uniformly until one of its dimensions (X or Y) fits the outside dimensions of the mask. Depending on the relative dimensions of the image source and mask, either the image source's width or height may be cropped or not fit the respective dimension of the mask.



Channel

The Channel menu determines the Channel of the input image used to create the mask. Choices include the red, green, blue, and alpha channels; the hue, luminance, or saturation values; or the auxiliary coverage channel of the input image (if one is provided).

Shadows/Midtones/Highlights

These buttons are used to select which range is output by the node as a mask. White pixels represent pixels that are considered to be part of the range, and black pixels are not included in the range. For example, choosing Shadows would show pixels considered to be shadows as white, and pixels that are not shadows as black. Mid gray pixels are only partly in the range and do not receive the full effect of any color adjustments to that range.

Channel

The Channel selection buttons shown in this tab can be used to extract a mask from the range of a specific color channel. By default, Fusion uses the luminance channel when the color ranges are examined.

Mini Spline Editor

The extent of the ranges is selected by manipulating the spline handles. There are four spline points, each with one Bézier handle. The two handles at the top represent the start of the shadow and highlight ranges; the two handles at the bottom represent the end of the range. The Bézier handles are used to control the falloff.

The midtones range has no specific control, since its range is understood to be the space between the shadow and highlight ranges. In other words, after low and high masks have been applied, midtones are everything else.

The X and Y text controls below the Mini Spline Editor can be used to enter precise positions for the selected Bézier point or handle.

Presets

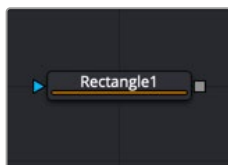
This sets the splines to two commonly-used configurations. The Simple button gives a straightforward linear-weighted selection, while the Smooth button uses a more natural falloff.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Rectangle Mask [Rec]



The Rectangle node

Rectangle Mask Node Introduction

The Rectangle mask creates simple square or rectangular masks. By default, it creates a rectangle in the same aspect ratio as the comp, but independent control is offered over the width, height, and angle, providing for a wide variety of rectangular shapes.

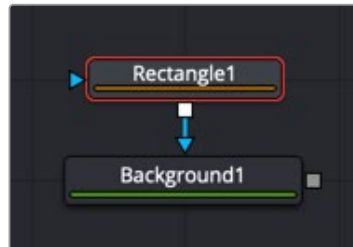
Inputs

The Rectangle mask node includes a single effect mask input.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

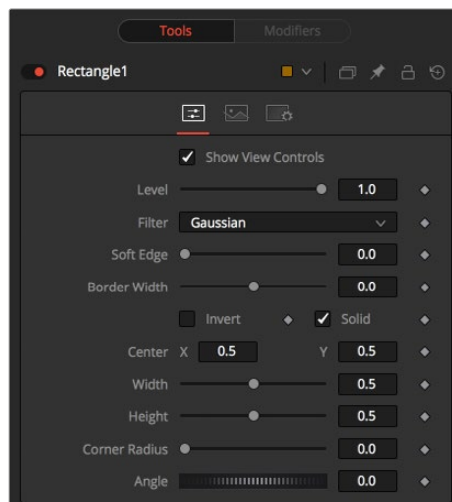
Basic Node Setup

The Rectangle mask node is useful for generating rectangular shapes. Below, the Rectangle mask is used to generate a colored square by cutting a square shape from a background node.



A Rectangle mask node creating a colored square by masking a Background node

Inspector



Rectangle Mask controls

Controls Tab

The Controls tab is used to refine how the rectangle appears after drawing it in the viewer.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the Blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Border Width

The Border Width control adjusts the thickness of the mask’s edge. When the solid checkbox is toggled on, the border thickens or narrows the mask. When the mask is not solid, the mask shape draws as an outline, and the width uses the Border Width setting.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask’s values add to the input mask’s values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask’s values.
- **Minimum:** Comparing the input mask’s values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask’s values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask’s values.
- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted: white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, this checkbox affects all pixels, regardless of whether the new mask covers them.

Solid

When the Solid checkbox is enabled, the mask is filled to be transparent (white) unless inverted. When disabled, the spline is drawn as just an outline whose thickness is determined by the Border Width slider.

Center X and Y

These controls adjust the position of the Rectangle mask.

Width and Height

Use these controls to change the X or Y scale of the rectangular effect mask independently of each other. Alternatively, drag the edges of the rectangle in the viewer to interactively adjust its size.

Corner Radius

Corner Radius allows the corners of the Rectangle mask to be rounded. A value of 0.0 is not rounding at all, which means that the rectangle has sharp corners. A value of 1.0 applies the maximum amount of rounding to the corners.

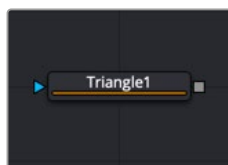
Angle

Change the rotation angle of an effect mask by moving the Angle control left or right. Values can be entered in the provided input boxes. Alternatively, use the onscreen controls by dragging the little circle at the end of the dashed angle line to interactively adjust the rotation of the ellipse.

Common Controls**Image and Settings Tabs**

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Triangle Mask [Tri]



The Triangle node

Triangle Mask Node Introduction

The Triangle mask creates simple triangular masks. It is unique in that it has no Center, Size, or Angle control. Unlike most other types of masks, all three points of the triangle can attach to a tracker or motion path.

Inputs

The Triangle mask node includes a single effect mask input.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

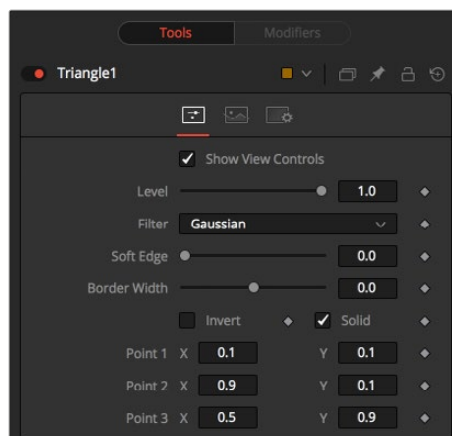
Basic Node Setup

The Triangle mask node is useful for generating triangular shapes. Below, the Triangle mask is used to generate a colored triangle by cutting a triangular shape from a background node.



A Triangle mask node creating a colored Triangle shape by masking a Background node

Inspector



Triangle Mask controls

Controls Tab

The Controls tab is used to refine how the triangle appears after drawing it in the viewer.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the Blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Border Width

The Border Width control adjusts the thickness of the mask’s edge. When the solid checkbox is toggled on, the border thickens or narrows the mask. When the mask is not solid, the mask shape draws as an outline, and the width uses the Border Width setting.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask’s values add to the input mask’s values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask’s values.
- **Minimum:** Comparing the input mask’s values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask’s values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask’s values.

- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted: white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, this checkbox affects all pixels, regardless of whether the new mask covers them.

Solid

When the Solid checkbox is enabled, the mask is filled to be transparent (white) unless inverted. When disabled, the spline is drawn as just an outline whose thickness is determined by the Border Width slider.

Point 1, Point 2, Point 3

These controls show the position coordinates of the three corners of the triangle. Each point can be published, connected to other controls, animated with a path, or attached to trackers. These tasks are performed by right-clicking the Position control or directly on the point in the viewer.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are also duplicated in other mask nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Wand Mask [Wnd]



The Wand node

Wand Mask Node Introduction

The Wand mask masks an image based on a wand-style selection, similar to the Magic Wand tool found in Adobe Photoshop. As with a Bitmap mask, any image in the composition can be the source of the mask. Generally, the default is most useful, where the source image is the input of the node to which the mask is applied.

When adding a Wand mask to a node, a crosshair appears in the viewers. This crosshair should be positioned in the image to select the color used to create the Wand mask. The mask is created by examining the pixel color beneath the selection point and adding that color to the mask. The mask then expands to examine the pixels surrounding the selection point. Surrounding pixels are added to the mask if they are the same color. The mask stops expanding when no connecting pixels fall within the color range of the mask.

Inputs

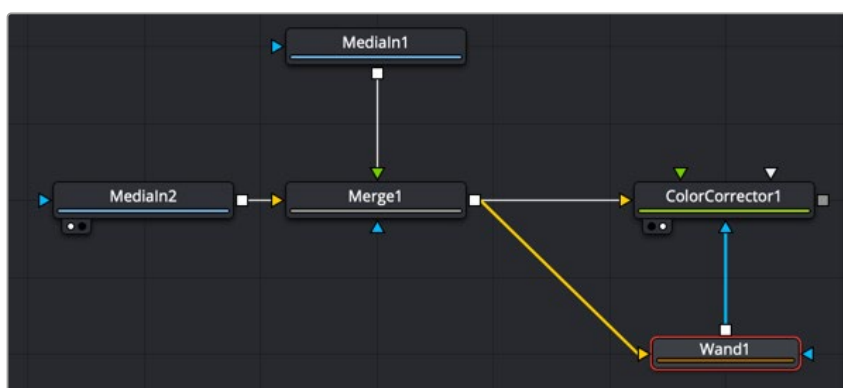
The Wand mask node includes two inputs in the Node Editor.

Input: The orange input accepts a 2D image from which the mask is created.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input combines the masks. How masks are combined is handled in the Paint mode menu in the Inspector.

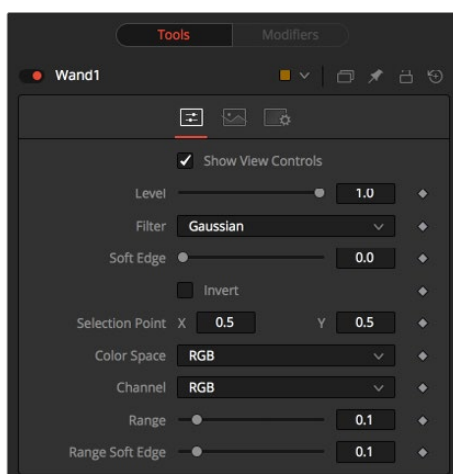
Basic Node Setup

The Wand mask node is not required for connecting an image into the effect mask input, but like the Bitmap node, it does provide options that are otherwise unavailable. It allows for selecting channels other than RGBA for the mask, as well as softness and clipping. In the node tree below, the Wand node takes the composite out of the merge, creating a mask for the color correction.



A Wand node selects a specific area in the image to create a mask.

Inspector



Wand Mask controls

Controls Tab

The Controls tab is used to refine how the mask appears after the Wand makes a selection in the viewer.

Show View Controls

The Show View Controls checkbox is used to enable/disable the display of the mask's onscreen controls in the viewer. Onscreen controls, including center position, polylines, angles, and others, do not appear when this checkbox is disabled, even when the node is selected.

Level

The Level control sets the transparency level of the pixels in the mask channel. When the value is 1.0, the mask is completely opaque (unless it has a soft edge). Lower values cause the mask to be partially transparent. The result is identical to lowering the Blend control of an effect.

NOTE: Lowering the level of a mask lowers the values of all pixels covered by the mask in the mask channel. For example, if a Circle mask is placed over a Rectangle mask, lowering the level of the Circle mask lowers the values of all the pixels in the mask channel, even though the Rectangle mask beneath it is still opaque.

Filter

This control selects the filtering algorithm used when applying Soft Edge to the mask.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Soft Edge

Use the Soft Edge slider to blur (feather) the mask, using the selected filter. Higher values cause the edge to fade off well beyond the boundaries of the mask. A value of 0.0 creates a crisp, well-defined edge.

Paint Mode

Connecting a mask to the effect mask input displays the Paint mode menu. The Paint mode is used to determine how the incoming mask for the effect mask input and the mask created in the node are combined.

- **Merge:** Merge is the default for all masks. The new mask is merged with the input mask.
- **Add:** The mask's values add to the input mask's values.
- **Subtract:** In the intersecting areas, the new mask values subtract from the input mask's values.
- **Minimum:** Comparing the input mask's values and the new mask, this displays the lowest (minimum) value.
- **Maximum:** Comparing the input mask's values and the new mask, this displays the highest (maximum) value.
- **Average:** This calculates the average (half the sum) of the new mask and the input mask.
- **Multiply:** This multiplies the values of the input mask by the new mask's values.

- **Replace:** The new mask completely replaces the input mask wherever they intersect. Areas that are zero (completely black) in the new mask do not affect the input mask.
- **Invert:** Areas of the input mask that are covered by the new mask are inverted; white becomes black and vice versa. Gray areas in the new mask are partially inverted.
- **Copy:** This mode completely discards the input mask and uses the new mask for all values.
- **Ignore:** This mode completely discards the new mask and uses the input mask for all values.

Invert

Selecting this checkbox inverts the entire mask. Unlike the Invert Paint mode, this checkbox affects all pixels, regardless of whether the new mask covers them.

Selection Point

The Selection Point is a pair of X and Y coordinates that determines where in the source image the Wand mask derives its initial color sample. This control is also seen as a crosshair in the viewers. The selection point can be positioned manually, connected to a tracker, path, or other expressions.

Color Space

The Color Space button group determines the color space used when selecting the source color for the mask. The Wand mask can operate in RGB, YUV, HLS, or LAB color spaces.

Channel

The Channel button group is used to select whether the color that is masked comes from all three color channels of the image, the alpha channel, or an individual channel only.

The exact labels of the buttons depend on the color space selected for the Wand mask operation. If the color space is RGB, the options are R, G, or B. If YUV is the color space, the options are Y, U, or V.

Range

The Range slider controls the range of colors around the source color that are included in the mask. If the value is left at 0.0, only pixels of the same color as the source are considered part of the mask. The higher the value, the more that similar colors in the source are considered to be wholly part of the mask.

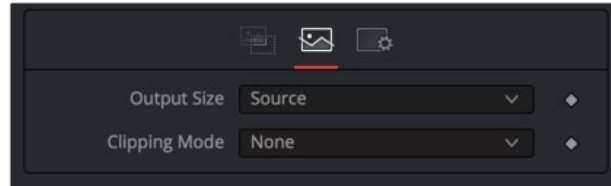
Range Soft Edge

The Range Soft Edge determines the falloff range of the colors selected. Any pixel within the range defined above are treated as 100% within the mask. If the soft range is set to 0.0, no other pixels are considered for the mask. Increasing the soft range increases the number of colors close to, but not quite within, the range included in the mask. These pixels are semitransparent in the mask.

The Common Controls

Nodes that create masks share several identical controls in the Inspector. This section describes controls that are common among mask nodes.

Inspector



Mask node Image tab

Image Tab

The controls in this tab set the resolution and clipping method used by the generated mask.

Output Size

The Output size menu sets the resolution of the mask node's output. The three options include the default resolution of the comp, the source input's resolution on nodes that have an input, or a custom resolution.

Custom

When selecting Custom from the Output Size menu, the width, height, and pixel aspect of the mask created are locked to values defined in the composition's Frame Format preferences. If the Frame Format preferences change, the resolution of the mask produced is changed to match. Disabling this option can be useful for building a composition at a different resolution than the eventual target resolution for the final render.

- **Width and Height:** This pair of controls is used to set the Width and Height dimensions of the mask to be created.
- **Pixel Aspect:** This control is used to specify the Pixel Aspect ratio of the created mask. An aspect ratio of 1:1 would generate a square pixel with the same dimensions on either side (like a computer monitor), and an aspect of 0.91 would create a slightly rectangular pixel (like an NTSC monitor).
- **Depth:** The Depth drop-down menu is used to set the pixel color depth of the image created by the mask. 32-bit pixels require four times the memory of 8-bit pixels but have far greater accuracy. Float pixels allow high dynamic range values outside the normal 0..1 range, for representing colors that are brighter than white or darker than black.

NOTE: Right-click on the Width, Height, or Pixel Aspect controls to display a menu listing the file formats defined in the preferences Frame Format tab. Selecting any of the listed options sets the width, height, and pixel aspect to the values for that format.

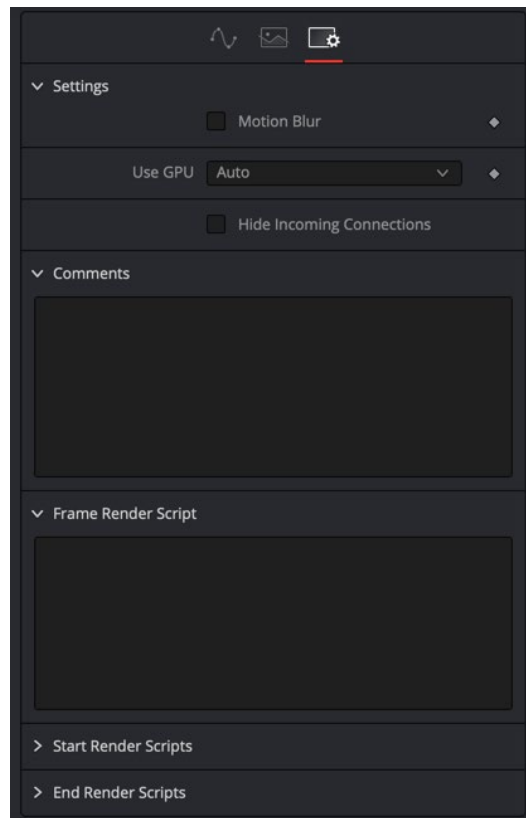
Clipping Mode

This option determines how the domain of definition rendering handles edges. The Clipping mode is most important when blur or softness is applied, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **None:** Setting this option to None does not perform any source image clipping. Any data required to process the node's effect that would usually be outside the upstream DoD is treated as black/transparent.

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Mask category. The Settings controls are even found on third-party plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options, which are also covered here.



Common Mask Settings controls

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware, and Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.